

```
import cv2
from google.colab.patches import cv2_imshow

# Read image in grayscale
img = cv2.imread("lab_image.jpg", 0)

# Check if image is loaded
if img is None:
    print("Image not found!")
else:
    # Apply Sobel operator along X-axis
    sobel_x = cv2.Sobel(img, cv2.CV_8U, 1, 0, ksize=5)

    # Save output image
    cv2.imwrite("sobel_x.jpg", sobel_x)

    # Display images
    print("Original Image")
    cv2_imshow(img)

    print("Sobel X Edge Detection")
    cv2_imshow(sobel_x)
```

Original Image



Sobel X Edge Detection

